

Microprocessor and Microcontroller

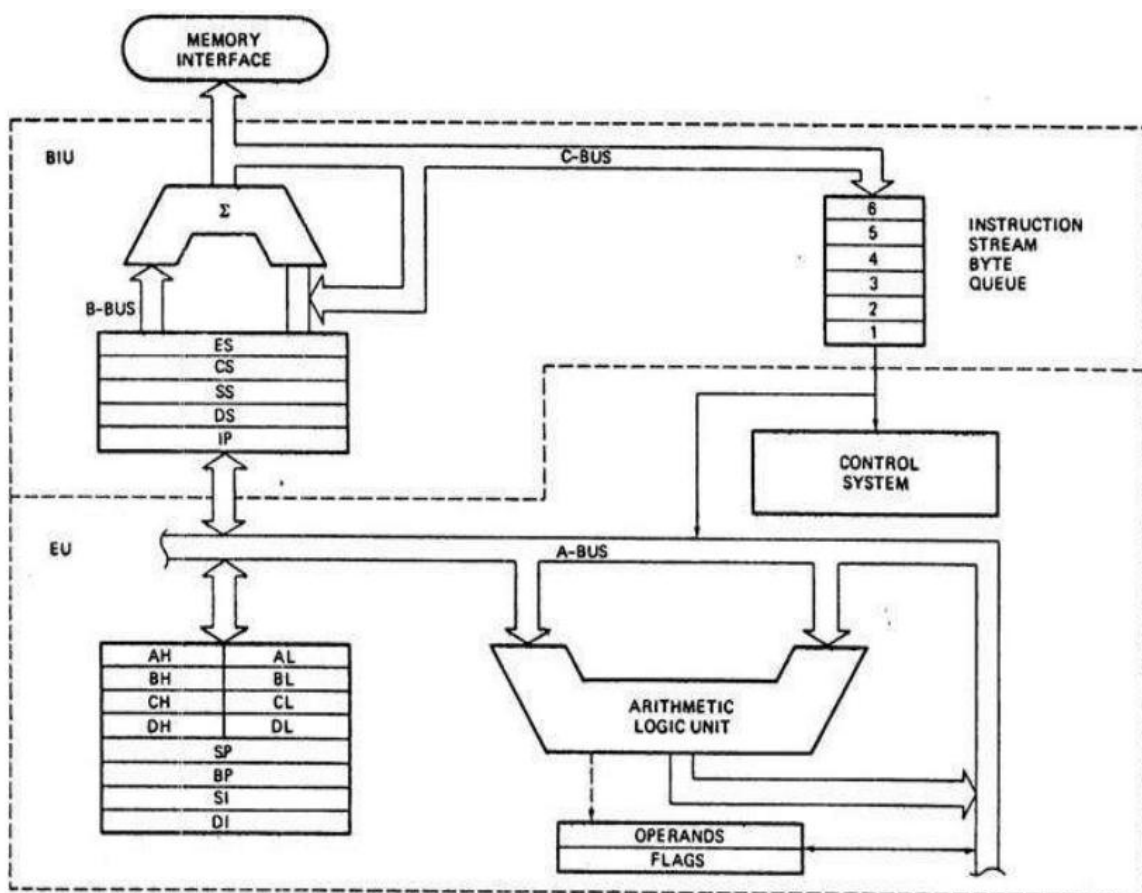
8086 ARCHITECTURE

Architecture of 8086 microprocessor:

As shown in the below figure, the 8086 CPU is divided into two independent functional parts

- Bus Interface Unit(BIU)
- Execution Unit(EU)

Dividing the work between these two units' speeds up processing.



The Execution Unit (EU):

The execution unit of the 8086 tells the BIU where to fetch instructions or data from, decodes instructions, and executes instructions. The EU contains control circuitry, which directs internal operations. A decoder in the EU translates instructions fetched from memory into a series of actions, which the EU carries out. The EU has a 16-bit arithmetic logic unit (ALU) which can add, subtract, AND, OR, XOR, increment, decrement, complement or shift binary numbers. The main functions of EU are:

- Decoding of Instructions

- Execution of instructions

Steps

1. EU extracts instructions from top of queue in BIU
2. Decode the instructions
3. Generates operands if necessary
4. Passes operands to BIU & requests it to perform read or write bus cycles to memory or I/O
5. Perform the operation specified by the instruction on operands

Bus Interface Unit (BIU):

- The BIU sends out addresses, fetches instructions from memory, reads data from ports and memory, and writes data to ports and memory.
- In simple words, the BIU handles all transfers of data and addresses on the buses for the execution unit.

8086 HAS PIPELINING ARCHITECTURE:

- While the EU is decoding an instruction or executing an instruction, which does not require use of the buses, the BIU fetches up to six instruction bytes for the following instructions.
- The BIU stores these pre-fetched bytes in a first-in-first-out register set called a queue.
- When the EU is ready for its next instruction from the queue in the BIU. This is much faster than sending out an address to the system memory and waiting for memory to send back the next instruction byte or bytes.
- Except in the case of JMP and CALL instructions, where the queue must be dumped and then reloaded starting from a new address, this pre-fetch and queue scheme greatly speeds up processing.
- Fetching the next instruction while the current instruction executes is called pipelining.

Register organization:

8086 has a powerful set of registers known as general purpose registers and special purpose registers. All of them are 16-bit registers.

General purpose registers:

These registers can be used as either 8-bit registers or 16-bit registers.

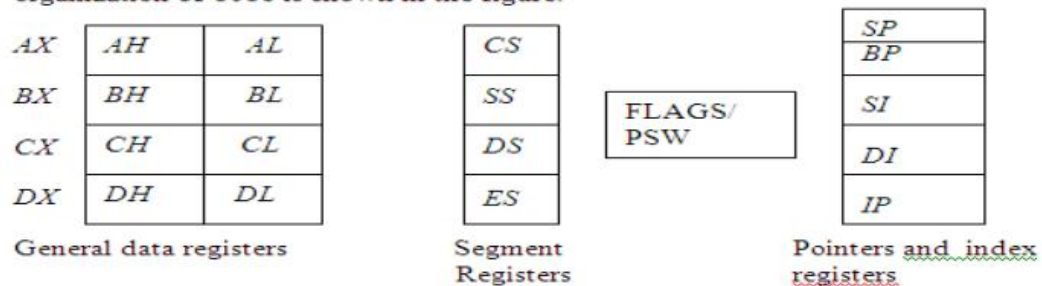
They may be either used for holding data, variables and intermediate results temporarily or for other purposes like a counter or for storing offset address for some particular addressing modes etc. Special purpose registers:

These registers are used as segment registers, pointers, index registers or as offset storage registers for particular addressing modes.

The 8086 registers are classified into the following types:

- General Data Registers
- Segment Registers
- Pointers and Index Registers
- Flag Register

The register set of 8086 can be categorized into 4 different groups. The register organization of 8086 is shown in the figure.



Register organization of 8086

8086 Architecture In Microprocessor And Microcontroller

MCQ QUESTIONS

1. The 8086 microprocessor is a ____ bit processor.

- A) 4-bit
 - B) 8-bit
 - C) 16-bit
 - D) 32-bit
-

2. The 8086 microprocessor has how many address lines?

- A) 16
 - B) 20
 - C) 24
 - D) 32
-

3. The memory segmentation in 8086 is done using ____ registers.

- A) General purpose
 - B) Segment
 - C) Pointer
 - D) Index
-

4. The 8086 microprocessor has how many general-purpose registers?

- A) 4
 - B) 6
 - C) 8
 - D) 10
-

5. The 8086 has a ____ address bus and a ____ data bus.

- A) 16-bit, 20-bit
 - B) 20-bit, 8-bit
 - C) 20-bit, 16-bit
 - D) 8-bit, 16-bit
-

6. The 8086 microprocessor has two functional units:

- A) ALU and Control Unit
 - B) Bus Interface Unit (BIU) and Execution Unit (EU)
 - C) Control Unit and Memory Unit
 - D) Decoder and Timer
-

7. The instruction queue in 8086 can hold how many bytes?

- A) 2
 - B) 4
 - C) 6
 - D) 8
-

8. The register that holds the offset address of the next instruction is:

- A) IP (Instruction Pointer)
 - B) SP (Stack Pointer)
 - C) BP (Base Pointer)
 - D) SI (Source Index)
-

9. Which register holds the base address of the current program?

- A) CS
 - B) DS
 - C) SS
 - D) ES
-

10. The maximum memory size that 8086 can address is:

- A) 64 KB
 - B) 256 KB
 - C) 512 KB
 - D) 1 MB
-

11. The 8086 uses segmented memory to:

- A) Increase processing speed
- B) Reduce hardware cost

- C) Access more than 64 KB memory using 16-bit registers
 - D) Simplify programming
-

12. The physical address in 8086 is calculated using:

- A) $\text{Segment} \times 16 + \text{Offset}$
 - B) $\text{Segment} + \text{Offset}$
 - C) $\text{Segment} \times 10 + \text{Offset}$
 - D) $\text{Segment} - \text{Offset}$
-

13. The flag register in 8086 is of size:

- A) 8 bits
 - B) 12 bits
 - C) 16 bits
 - D) 32 bits
-

14. Which flag is set when the result of an operation is zero?

- A) Carry Flag
 - B) Zero Flag
 - C) Sign Flag
 - D) Overflow Flag
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15. Which of the following is *not* a segment register in 8086?

- A) CS
- B) DS
- C) ES
- D) PSW